



Photo taken with Prof Nazri

SUMMARY OF THE VISIT

The industrial visit to Universiti Tun Hussein Onn Malaysia (UTHM) gave us a clear view of how the institution has evolved digitally. We explored the history of the university, starting with a polytechnic training centre for staffs and then becoming an independent entity, which required a large upgrade to its ICT capabilities. The core focus of the visit is Enterprise Architecture of the institution. We also learnt about the strategic consistency of information technology with the university's mission, especially the formulation of the ICT Strategic Plan 2026-2030. We learned how important it is to align IT planning with the university's mission. Prof. Nazri also connected physical infrastructure such as the Tier 3 data center design with the necessary governance to manage digital transformation effectively in a large academic ecosystem.



Group photo taken at the main entrance of UTHM Digital Gateway, Batu Pahat.

TECHNOLOGIES AND ISSUES DISCUSSED

1. We learned about the **specific cooling technologies** used in the data centre to maintain sustainability. The facility uses Hot Aisle Containment and Cold Aisle Containment to keep the server room temperature between 25-27°C without wasting energy. Mr. Norashid also highlighted a regional issue where water shortages in Singapore are driving data centers to move to Johor. This has led to a shift to liquid cooling technologies to protect natural resources while supporting high-density customers like Google and AWS.
2. **Network security** is another major topic discussed during the visit. The data centre uses double-firewall architecture and a web-based security wall to block attacks from frequent regions like Europe and China. Public universities (UA) do not use commercial public clouds, but rely on government-supported Private Data Centre (PDSA) and Disaster Recovery Center (DRC) in Pagoh to make sure data sovereignty and security.
3. We discussed the **challenges of digital transformation and user experience**. The IT team is currently migrating from old legacy client-server systems to modern web-based platforms, and this complex process took more than three years to reach 70 - 80% completion. We also know that the most common user complaints is not technical failure, but forgetting passwords due to the Single Sign-On (SSO) implementation.

REFLECTION | CHEW CHIU XIAN (A23CS0061)

The UTHM visit has helped me to understand how the Enterprise Architecture in UTHM functions. Firstly, I learned about two types of containment in data centre, which are Hot and Cold Aisle Containment and how they work in practice, with the purpose of reducing power consumption, while Hot Aisle Containment is in hot temperature. Besides that, the double-firewall architecture and the decision to use the Government Private Data Centre (PDSA) instead of public clouds, taught me the importance of data sovereignty in the public sector.

REFLECTION | LAU YAN KAI (A23CS0098)

From this visit, I learnt that technology cannot succeed without considering human behaviour because even with advanced systems like Single Sign-on (SSO) are used, human errors like forgetting passwords are still a major issue. Besides, I was very impressed by the Tier 3 Data Center and the In-Row Cooling engineering. The discussion on water cooling taught me a valuable lesson about sustainability. It shows that innovation is not just about being fast, but it is about designing systems that protect our natural resources for the future.